

UNIT 4: GLOBAL SUSTAINABLE AGRICULTURE

Class Time: Minimum 75 Minutes

Description

This lesson will encourage the students to work with G.I.S in particular the ArcView software to think critically about the issue of sustainability throughout the world. In this lesson the students will:

- Explore world agriculture in relation to country size, arable land and population.
- Prepare maps of country size, arable land and agricultural production.
- Analyze the percentage of arable land worldwide and determine the relationship with agricultural production.
- Use and query data tables linked and associated with maps.
- Compute and use new data tables.
- Display computed data using charts.
- Prepare a map layout for printing.

Curriculum Expectations

Overall Expectations:

HEV.02 – analyse geographic issues that arise from the impact of human activities on the environmentally different regions of the world.

HEV.03 – evaluate approaches, policies and principals relating to the protection and sustainability of the planet's life support system.

UCV.01 – analyse, forecast, and evaluate changes in the human use of the earth and its resources.

Specific Expectations:

HE1.01 – demonstrate an understanding of how human – induced changes in natural systems can diminish their capacity for supporting human activity.

HE2.02 – analyse the impact of past and current trends in agriculture (e.g., Green Revolution, corporate farming, biotechnology, monoculture, organic farming) on natural and human systems.

HE2.04 – evaluate the economic, social, and ecological impact of current methods for raising or harvesting a selected resource.

GI1.02 – demonstrate an understanding of the technologies used in the analysis and synthesis of geographic data (e.g., remote sensing, information technology, geographic information system, hypermedia).

GI2.02 – analyse cause and effect and sequence relationships in geographical data.

GI2.06 – draw conclusions or make judgements or prediction on the basis of reasoned analysis.

GI2.07 – produce mental, thematic, and choropleth maps to illustrate local or global patterns.

GI3.01 – use different types of maps and images (e.g., topographic maps, thematic maps, satellite imagery) to analyse the consequence of human activities or environmental phenomena.

Prior Knowledge and Skills

- Basic knowledge of the ArcView Software (See Power Point Presentation ‘The Basic of ArcView’).
- Knowledge of global environmental trends (climate change, biodiversity, pollution, desertification, deforestation) and a familiarisation with the international agreements that address these trends (Kyoto protocol, the Convention on Biodiversity, etc).
- An understanding of human development trends and how they influence geographic issues (birth/death rates, GDP, child mortality, emissions, etc).
- Statistical Analysis, Geographical Information System (GIS).

Planning Notes

- You will need a computer system with ArcView GIS 3.2 software to carry out this unit.
- Download the data from the website <http://www.ocdsb.edu.on.ca/woodsweb/Geomatics> before doing the lesson.
- Photocopy of Worksheet 12 Global Sustainable Agriculture.

Teaching/Learning Strategies

Most of the learning strategies are discussed in each activity, but here are some of the overall strategies for the unit.

- Students should do this unit with a partner.
- Introduce Arcview GIS 3.2 to the students with Power Point Presentation “The Basis of Arcview”.
- Make a printout of Worksheet 12: Global Sustainable Agriculture, and distribute it to every group.
- Explain to the students that they will have to produce a Global Sustainable Agriculture World Map. Their map will be evaluated using cartographic skills and a rubric evaluation included in this unit.
- Assign students the suggested homework where students are encouraged to do an in-depth analysis of an agricultural issue to determine its implications for sustainable development.
- Introduce students to the new vocabulary and concepts listed in the activity. Use the glossary at the end of the sustainable development module.

Accommodations

Frequent monitoring and assistance should be provided. Accommodations that are normally provided in the regular classroom for students with special needs should be provided. Here are some strategies for helping them.

- Information should be presented to as many modalities as possible through oral presentations, board notes, diagrams, class discussions, activity-based learning.
- Instructions should be clear, concise, to the point, verbally rehearsed by the student, properly sequenced, reviewed after a time lapse, written down on the board or in a notebook.
- Assignments should be outlined in writing, written in the students’ homework/assignment book, written (and left) on the blackboard, broken down by step with the teacher, rehearsed / modelled, and scheduled step by step.
- Questioning should be clear, concise, allowing for appropriate response time. Students need time to organise their thoughts, match them with the appropriate vocabulary, put them in correct syntax, sequence the ideas and finally express them with good speech. All these skills, most others take for granted.
- Encourage students to question for clarification and additional information. Learn to speak at a slower rate of speech. Avoid figurative language unless it has been specifically pre-taught. Encourage and evaluate oral responses and class participation. Allow extra time (e.g. homework) for students to complete assignments that others students may finish in class time.

Create extra practice opportunities with new vocabulary or concepts. Provide overviews or lesson highlights at the beginning of class. Discuss and simplify sophisticated vocabulary or language. Finally, give several short assignments rather than one long one.

- Provide photocopies of the GIS lesson and the Background Information for students with special needs instead of referring them to the on-line version.